

Queue

Question No:1

DS

Session

Queue

Question Information

Level 1 Challenge 51

Problem Description

The Monk recently learnt about priority queues and requested his teacher for a fun challenge to solve. As a result, his teacher devised a simple task. He now possesses A , which is an integer array. He wishes to discover the product of the greatest, second largest, and third largest integers in the range $[1, i]$ for each index i .

Note: Two numbers can be the same value-wise but they should be distinct index-wise.

Constraints:

$1 \leq N \leq 100000$

$0 \leq A[i] \leq 1000000$

Input:

The first line contains an integer N , denoting the number of elements in the array A .

The next line contains N space separated integers, each denoting the i th integer of the array A .

Output:

Print the answer for each index in each line. If there is no second largest or third largest number in the array A upto that index, then print "-1", without the quotes.

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
1 2 3 4 5

EXPECTED OUTPUT

-1
-1
6
24
60

Test Case 2

INPUT (STDIN)

8
3 7 1 2 4 7 5 9

EXPECTED OUTPUT

-1
-1
21
42
84
196
245
441

Code:

```
1  #include <stdio.h>
2  int main()
3  {
4      int N;
5      scanf("%d", &N);
6
7      int A[N];
8
9      for (int i = 0; i < N; i++)
10     {
11         scanf("%d", &A[i]);
12     }
13
14     int first = -1, second = -1, third = -1;
15
16     for (int i = 0; i < N; i++)
17     {
18         int x = A[i];
19
20         if (x >= first)
21         {
22             third = second;
23             second = first;
24             first = x;
25         }
26         else if (x >= second)
27         {
28             third = second;
29             second = x;
30         }
31         else if (x >= third)
32         {
33             third = x;
34         }
35
36         if (i < 2)
37         {
38             printf("-1\n");
39         }
40     }
41     else
42     {
43         long long product = (long long)first * second * third;
44         printf("%lld\n", product);
45     }
46
47     return 0;
48 }
```

Output:

Test Case-1:

```
5
1 2 3 4 5
-1
-1
6
24
60

-----
Process exited after 13.52 seconds with return value 0
Press any key to continue . . . |
```

Test Case-2:

```
8
3 7 1 2 4 7 5 9
-1
-1
21
42
84
196
245
441

-----
Process exited after 14.3 seconds with return value 0
Press any key to continue . . . |
```